

SCOPE

VLANs

10–100 + native 99

Access ports

ASW1–ASW6 · IDF A/B/C

PortFast + BPDU Guard

Fa0/1–20 end-device ports

802.1Q trunks

native 99

L3 EtherChannel

Po1 · LACP · /30

SVIs

L3SW1 .2 · L3SW2 .3

Floor plan

MDF / IDF mapping

BANK FACILITY NETWORK DESIGN

Bank Network

Layer 2 & Layer 3

Configuration Commands

Configuration command book · Cisco Packet Tracer · Bank Main Branch

AUTHOR

Nondumiso Mbuyazi

Layer 2 & Layer 3 (switching, SVIs, L3 EtherChannel)

HSRP, DHCP, DNS, routing: Innocent Nkosiye Mbatha — see Bank L3 & L7 Configuration Report

Project: Bank Facility Network Design · Revision 1.0 · 30 September 2026 · Internal / Portfolio

Sources: live port table, 2026-09-16 live extract, L3SW1/L3SW2 CLI captures (BANK-live-2026-09-16/18.pkt)

1

Device & Addressing Reference

Scope: L2 + base L3 (switching, SVIs, Po1)

Device	Role / location	Trunk ports	Other ports	Addressing / VLANs
L3SW1 (DSW1)	Distribution, MDF	Gi1/0/3–8 → ASW1–ASW6	Gi1/0/23–24 = Po1	SVIs .2 • Po1 10.255.255.1/30
L3SW2 (DSW2)	Distribution, MDF	Gi1/0/3–8 → ASW1–ASW6	Gi1/0/23–24 = Po1	SVIs .3 • Po1 10.255.255.2/30
ASW1	Access, IDF A	Gi0/1 → L3SW1, Gi0/2 → L3SW2 (Dist Gi1/0/3)	Fa0/1–17	VLANs 20, 30, 40, 60
ASW2 (PoE)	Access, IDF A	Gi0/1 → L3SW1, Gi0/2 → L3SW2 (Dist Gi1/0/4)	Fa0/1–15	VLANs 20, 30, 40, 100, 60
ASW3	Access, IDF B	Gi0/1 → L3SW1, Gi0/2 → L3SW2 (Dist Gi1/0/5)	Fa0/1–6	VLANs 70, 40
ASW4 (PoE)	Access, IDF B	Gi0/1 → L3SW1, Gi0/2 → L3SW2 (Dist Gi1/0/6)	Fa0/1–6	VLANs 70, 40
ASW5	Access, IDF C	Gi0/1 → L3SW1, Gi0/2 → L3SW2 (Dist Gi1/0/7)	Fa0/1–15	VLANs 50, 10, 80, 90
ASW6 (PoE)	Access, IDF C	Gi0/1 → L3SW1, Gi0/2 → L3SW2 (Dist Gi1/0/8)	Fa0/1–13	VLANs 10, 50, 80, 90, 100

Addressing rules

Item	Value
User subnets	10.10.<vlan>.0/24 (VLANs 10–100)
L3SW1 SVI	10.10.<vlan>.2
L3SW2 SVI	10.10.<vlan>.3
Gateway .1	HSRP virtual IP — never an SVI address
Native VLAN 99	Trunks only: no SVI, no IP, no users
Po1	10.255.255.0/30 — L3SW1 .1, L3SW2 .2

Ownership

Area	Owner
VLANs, access ports, VLAN colours, end devices, floor plan	Nondumiso Mbuyazi
Trunks, native 99, PortFast, BPDU Guard	Nondumiso Mbuyazi
L3 EtherChannel Po1 (LACP), SVIs + ip routing	Nondumiso Mbuyazi
HSRP, DHCP + helpers, DNS, static routing	Innocent Nkosiye Mbathe — see <i>Bank L3 & L7 Configuration Report</i>

NOTE

Device names follow the live topology: **L3SW1** / **L3SW2** (DSW1 / DSW2 in the original script). Edge links to Router0/Router1 are routed ports owned by the L3 report and are not touched here.

2

VLAN Database

Create on every switch

VLAN	Name	Subnet	Canvas colour	Access switches
10	MANAGEMENT	10.10.10.0/24	White	ASW5, ASW6
20	FINANCE	10.10.20.0/24	Pink	ASW1, ASW2
30	HR	10.10.30.0/24	Red	ASW1, ASW2
40	IT	10.10.40.0/24	—	ASW1, ASW2, ASW3, ASW4
50	TELLERS	10.10.50.0/24	Yellow	ASW5, ASW6
60	CUSTOMER_SERVICE	10.10.60.0/24	Green	ASW1, ASW2
70	LOANS	10.10.70.0/24	Blue	ASW3, ASW4
80	SECURITY	10.10.80.0/24	Peach	ASW5, ASW6
90	STORAGE_SERVERS	10.10.90.0/24	—	ASW5, ASW6
99	NATIVE	—	—	Trunk native only — no users, no SVI
100	GUEST	10.10.100.0/24	—	ASW2, ASW6

WARNING

VLAN 99 is the trunk native VLAN only. The script's VLAN 999 NETWORK_MANAGEMENT (10.10.99.0/24, interface `vlan 999`) belongs to a separate campus drawing and has no bank equivalent — omitted.

NOTE

The Bank L3 & L7 Configuration Report uses short names for the same VLAN IDs: 60 CustomerSVC, 90 Servers, 100 **Wireless** (= GUEST here), plus Management/Finance/... in mixed case.

VLAN creation — all switches

L2

```
! ALL switches: L3SW1, L3SW2, ASW1-ASW6
enable
configure terminal
vlan 10
  name MANAGEMENT
vlan 20
  name FINANCE
vlan 30
  name HR
vlan 40
  name IT
vlan 50
  name TELLERS
vlan 60
  name CUSTOMER_SERVICE
vlan 70
  name LOANS
vlan 80
  name SECURITY
vlan 90
  name STORAGE_SERVERS
vlan 99
  name NATIVE
vlan 100
  name GUEST
end
write memory
```

3

Access Ports, End Devices & Floor Plan

switchport mode access • IDF A / B / C

Switch	IDF	Ports	VLAN	Name
ASW1	IDF A	Fa0/1 - 3	20	FINANCE
ASW1	IDF A	Fa0/4 - 5	30	HR
ASW1	IDF A	Fa0/6 - 9	40	IT
ASW1	IDF A	Fa0/15 - 17	60	CUSTOMER_SERVICE
ASW2 (PoE)	IDF A	Fa0/1 - 3	20	FINANCE
ASW2 (PoE)	IDF A	Fa0/4 - 6	30	HR
ASW2 (PoE)	IDF A	Fa0/7 - 9	40	IT
ASW2 (PoE)	IDF A	Fa0/10 - 12	100	GUEST
ASW2 (PoE)	IDF A	Fa0/13 - 15	60	CUSTOMER_SERVICE
ASW3	IDF B	Fa0/1 - 4	70	LOANS
ASW3	IDF B	Fa0/5 - 6	40	IT
ASW4 (PoE)	IDF B	Fa0/1 - 3	70	LOANS
ASW4 (PoE)	IDF B	Fa0/4 - 6	40	IT
ASW5	IDF C	Fa0/1 - 5	50	TELLERS
ASW5	IDF C	Fa0/6 - 9	10	MANAGEMENT
ASW5	IDF C	Fa0/10 - 11	80	SECURITY
ASW5	IDF C	Fa0/12 - 13	90	STORAGE_SERVERS
ASW5	IDF C	Fa0/14 - 15	80	SECURITY
ASW6 (PoE)	IDF C	Fa0/1 - 3	10	MANAGEMENT
ASW6 (PoE)	IDF C	Fa0/4 - 5	50	TELLERS
ASW6 (PoE)	IDF C	Fa0/6 - 8	80	SECURITY
ASW6 (PoE)	IDF C	Fa0/9 - 11	90	STORAGE_SERVERS
ASW6 (PoE)	IDF C	Fa0/12 - 13	100	GUEST

Floor plan / IDF mapping

Location	Switches	VLANs served
MDF — Network Room (Room 12)	L3SW1, L3SW2 (+ Router0/1)	Distribution; fibre backbone to wall racks WR-01...WR-04
IDF A	ASW1, ASW2 (PoE)	20, 30, 40, 60, 100
IDF B	ASW3, ASW4 (PoE)	40, 70
IDF C	ASW5, ASW6 (PoE)	10, 50, 80, 90, 100

Differences vs earlier port table — this book follows her script

Item	Earlier port table (VLANs)	This book (VLANs)
ASW1	50 (Fa0/1-8), 80 (9-11), 60 (12)	20, 30, 40, 60 (Fa0/15-17)
ASW2	60 (Fa0/1-6), 20 (7-10), 70 laptops	20, 30, 40, 100, 60
ASW3	10, 30, 20, 70 (ports Fa0/?)	70, 40
ASW4	20 + IP Phone1, 30, 70	70, 40
ASW5	80 (PC42-46), 20 laptops	50, 10, 80, 90, 80
ASW6	90 servers, 80 / 40	10, 50, 80, 90, 100
VLAN 100	APs on ASW7 only	Also ASW2 Fa0/10-12, ASW6 Fa0/12-13
Dist / edge	Dist Gi1/0/? (verify); per-port PortFast	Gi1/0/3-8 fixed map; PortFast range Fa0/1-20

NOTE

Also in live topology: ASW7 with 12 access points on VLAN 100 (Fa0/1-12). Not part of this script — if ASW7 is kept, repeat the same VLAN, access, trunk and PortFast steps for it.

3 Access Port Commands — ASW1 to ASW3

IDF A + IDF B

ASW1 — IDF A

IDF A

```
! ASW1 — IDF A
enable
configure terminal
hostname ASW1
!
interface range fa0/1 - 3
 switchport mode access
 switchport access vlan 20
!
interface range fa0/4 - 5
 switchport mode access
 switchport access vlan 30
!
interface range fa0/6 - 9
 switchport mode access
 switchport access vlan 40
!
interface range fa0/15 - 17
 switchport mode access
 switchport access vlan 60
!
! PortFast + BPDU Guard - end-device ports
interface range fa0/1 - 20
 spanning-tree portfast
 spanning-tree bpduguard enable
!
end
write memory
```

ASW2 — IDF A (PoE)

IDF A • PoE

```
! ASW2 — IDF A (PoE)
enable
configure terminal
hostname ASW2
!
interface range fa0/1 - 3
 switchport mode access
 switchport access vlan 20
!
interface range fa0/4 - 6
 switchport mode access
 switchport access vlan 30
!
interface range fa0/7 - 9
 switchport mode access
 switchport access vlan 40
!
interface range fa0/10 - 12
 switchport mode access
 switchport access vlan 100
!
interface range fa0/13 - 15
 switchport mode access
 switchport access vlan 60
!
! PortFast + BPDU Guard - end-device ports
interface range fa0/1 - 20
 spanning-tree portfast
 spanning-tree bpduguard enable
!
end
write memory
```

ASW3 — IDF B

IDF B

```
! ASW3 — IDF B
enable
configure terminal
hostname ASW3
!
interface range fa0/1 - 4
 switchport mode access
 switchport access vlan 70
!
interface range fa0/5 - 6
 switchport mode access
 switchport access vlan 40
!
! PortFast + BPDU Guard - end-device ports
interface range fa0/1 - 20
 spanning-tree portfast
 spanning-tree bpduguard enable
!
end
write memory
```

3 Access Port Commands — ASW4 to ASW6

IDF B + IDF C

ASW4 — IDF B (PoE)

IDF B • PoE

```
! ASW4 — IDF B (PoE)
enable
configure terminal
hostname ASW4
!
interface range fa0/1 - 3
 switchport mode access
 switchport access vlan 70
!
interface range fa0/4 - 6
 switchport mode access
 switchport access vlan 40
!
! PortFast + BPDU Guard - end-device ports
interface range fa0/1 - 20
 spanning-tree portfast
 spanning-tree bpduguard enable
!
end
write memory
```

ASW5 — IDF C

IDF C

```
! ASW5 — IDF C
enable
configure terminal
hostname ASW5
!
interface range fa0/1 - 5
 switchport mode access
 switchport access vlan 50
!
interface range fa0/6 - 9
 switchport mode access
 switchport access vlan 10
!
interface range fa0/10 - 11
 switchport mode access
 switchport access vlan 80
!
interface range fa0/12 - 13
 switchport mode access
 switchport access vlan 90
!
interface range fa0/14 - 15
 switchport mode access
 switchport access vlan 80
!
! PortFast + BPDU Guard - end-device ports
interface range fa0/1 - 20
 spanning-tree portfast
 spanning-tree bpduguard enable
!
end
write memory
```

ASW6 — IDF C (PoE)

IDF C • PoE

```
! ASW6 — IDF C (PoE)
enable
configure terminal
hostname ASW6
!
interface range fa0/1 - 3
 switchport mode access
 switchport access vlan 10
!
interface range fa0/4 - 5
 switchport mode access
 switchport access vlan 50
!
interface range fa0/6 - 8
 switchport mode access
 switchport access vlan 80
!
interface range fa0/9 - 11
 switchport mode access
 switchport access vlan 90
!
interface range fa0/12 - 13
 switchport mode access
 switchport access vlan 100
!
! PortFast + BPDU Guard - end-device ports
interface range fa0/1 - 20
 spanning-tree portfast
 spanning-tree bpduguard enable
!
end
write memory
```

4 802.1Q Trunks, PortFast & BPDU Guard

Native VLAN 99 on every trunk

Access switches — uplinks (ASW1-ASW6)

ALL ASW

```
! ASW1-ASW6: Gi0/1 + Gi0/2 = redundant uplinks
! Gi0/1 -> L3SW1, Gi0/2 -> L3SW2
interface range gi0/1 - 2
  description UPLINK to L3SW1 / L3SW2
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,99,100
  switchport nonegotiate
  no shutdown
!
end
write memory
```

Distribution — downlinks (L3SW1 + L3SW2)

BOTH DIST

```
! L3SW1 and L3SW2 (identical)
! Gi1/0/3 ASW1 | Gi1/0/4 ASW2 | Gi1/0/5 ASW3
! Gi1/0/6 ASW4 | Gi1/0/7 ASW5 | Gi1/0/8 ASW6
! Never Gi1/0/1 (routed edge to Router0)
interface range gigabitEthernet1/0/3 - 8
  description DOWNLINK to ASW1-ASW6
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,99,100
  switchport nonegotiate
  no shutdown
!
end
write memory
```

Adding a VLAN to a trunk later

```
! Later, to ADD one VLAN to an existing trunk:
interface range gi0/1 - 2
  switchport trunk allowed vlan add 110
!
! WRONG (replaces the whole list = outage):
! switchport trunk allowed vlan 110
```

PortFast + BPDU Guard rules

EDGE ONLY

```
! Access switches ASW1-ASW6
! (included in every block in section 3)
interface range fa0/1 - 20
  spanning-tree portfast
  spanning-tree bpduguard enable
!
! Fa0/1-20 = end-device ports only.
! NEVER on trunks / uplinks:
!   ASW Gi0/1-2, Dist Gi1/0/3-8, Po1
!
! Recover a port that BPDU Guard err-disabled
! (remove the rogue switch first):
interface fastEthernet0/5
  shutdown
  no shutdown
```

WARNING

Always use `switchport trunk allowed vlan add` to extend a trunk. Without **add** the command replaces the list and every other VLAN drops off the trunk.

NOTE

Native VLAN and allowed list must match on both ends of every trunk. VLAN 99 is included in the allowed list so the native VLAN is carried consistently.

5 Layer 3 EtherChannel — Po1 (LACP)

Gi1/0/23-24 • 10.255.255.0/30

L3SW1 — Po1 10.255.255.1/30

LACP ACTIVE

```
! L3SW1 — Po1 routed link to L3SW2
enable
configure terminal
interface range gigabitEthernet1/0/23 - 24
description Po1 member to L3SW2
no switchport
channel-group 1 mode active
no shutdown
!
interface port-channel 1
description L3 EtherChannel to L3SW2
no switchport
ip address 10.255.255.1 255.255.255.252
no shutdown
!
end
write memory
```

L3SW2 — Po1 10.255.255.2/30

LACP ACTIVE

```
! L3SW2 — Po1 routed link to L3SW1
enable
configure terminal
interface range gigabitEthernet1/0/23 - 24
description Po1 member to L3SW1
no switchport
channel-group 1 mode active
no shutdown
!
interface port-channel 1
description L3 EtherChannel to L3SW1
no switchport
ip address 10.255.255.2 255.255.255.252
no shutdown
!
end
write memory
```

Check	Rule
Mode	LACP active on both sides
Members	Same speed/duplex, both no switchport before bundling
Addressing	IP only on Port-channel1, never on member ports
Live state	L3SW2 capture shows Port-channel1 10.255.255.2 up/up (routed) — matches this design
Members in PT	Gi1/0/23-24 not visible in the live capture — confirm with show etherchannel summary
Effect	Dist-to-Dist link is routed: VLAN traffic between L3SW1/L3SW2 crosses the ASW trunks

6 Switched Virtual Interfaces (SVIs)

L3SW1 .2 • L3SW2 .3 • .1 reserved for HSRP

L3SW1 — SVIs (.2)

DSW1

```
! L3SW1 — SVIs use .2 (.1 = HSRP VIP)
configure terminal
ip routing
!
interface vlan 10
 ip address 10.10.10.2 255.255.255.0
 no shutdown
interface vlan 20
 ip address 10.10.20.2 255.255.255.0
 no shutdown
interface vlan 30
 ip address 10.10.30.2 255.255.255.0
 no shutdown
interface vlan 40
 ip address 10.10.40.2 255.255.255.0
 no shutdown
interface vlan 50
 ip address 10.10.50.2 255.255.255.0
 no shutdown
interface vlan 60
 ip address 10.10.60.2 255.255.255.0
 no shutdown
interface vlan 70
 ip address 10.10.70.2 255.255.255.0
 no shutdown
interface vlan 80
 ip address 10.10.80.2 255.255.255.0
 no shutdown
interface vlan 90
 ip address 10.10.90.2 255.255.255.0
 no shutdown
interface vlan 100
 ip address 10.10.100.2 255.255.255.0
 no shutdown
!
! No SVI for VLAN 99 (native only)
end
write memory
```

L3SW2 — SVIs (.3)

DSW2

```
! L3SW2 — SVIs use .3 (.1 = HSRP VIP)
configure terminal
ip routing
!
interface vlan 10
 ip address 10.10.10.3 255.255.255.0
 no shutdown
interface vlan 20
 ip address 10.10.20.3 255.255.255.0
 no shutdown
interface vlan 30
 ip address 10.10.30.3 255.255.255.0
 no shutdown
interface vlan 40
 ip address 10.10.40.3 255.255.255.0
 no shutdown
interface vlan 50
 ip address 10.10.50.3 255.255.255.0
 no shutdown
interface vlan 60
 ip address 10.10.60.3 255.255.255.0
 no shutdown
interface vlan 70
 ip address 10.10.70.3 255.255.255.0
 no shutdown
interface vlan 80
 ip address 10.10.80.3 255.255.255.0
 no shutdown
interface vlan 90
 ip address 10.10.90.3 255.255.255.0
 no shutdown
interface vlan 100
 ip address 10.10.100.3 255.255.255.0
 no shutdown
!
! No SVI for VLAN 99 (native only)
end
write memory
```

WARNING

No SVI may use .1. 10.10.<vlan>.1 is the HSRP virtual gateway configured by Nkosiyethu. The 2026-09-16 capture showed L3SW2 Vlan60 = 10.10.60.1 — the cause of TKT-2109. Correct value: 10.10.60.3.

VLAN	L3SW1	L3SW2	VIP (HSRP)
10	10.10.10.2	10.10.10.3	10.10.10.1
20	10.10.20.2	10.10.20.3	10.10.20.1
30	10.10.30.2	10.10.30.3	10.10.30.1
40	10.10.40.2	10.10.40.3	10.10.40.1
50	10.10.50.2	10.10.50.3	10.10.50.1
60	10.10.60.2	10.10.60.3	10.10.60.1
70	10.10.70.2	10.10.70.3	10.10.70.1
80	10.10.80.2	10.10.80.3	10.10.80.1
90	10.10.90.2	10.10.90.3	10.10.90.1
100	10.10.100.2	10.10.100.3	10.10.100.1

HSRP standby lines, DHCP helpers and routes are added on top of these SVIs — see *Bank L3 & L7 Configuration Report*, sections 3–5.

7 Configuration Order & Verification

Expected results — capture real output for evidence

Configuration order

1	VLANs on every switch (section 2).
2	Trunks — Dist downlinks, then ASW uplinks; same native 99 and allowed list both ends (section 4).
3	Access ports + PortFast/BPDU Guard (Fa0/1–20) per ASW (section 3).
4	Po1 L3 EtherChannel between L3SW1 and L3SW2 (section 5).
5	SVIs + ip routing on L3SW1 (.2) and L3SW2 (.3) (section 6).
6	Hand-over to Nkosiyeu: HSRP, DHCP helpers, static routes (Bank L3 & L7 Configuration Report).
7	Verify with the table on this page, then write memory on every device.

Open L2 items (verify in live .pkt)

Port assignments in this book differ from the earlier port table (section 3) — confirm the running .pkt matches this book before closing evidence.

ASW7 (12 APs, VLAN 100) is in the live topology but not in this script — if kept, repeat sections 2–4 for it (L3SW2 capture shows Gi1/0/9 up).

L3SW2 Vlan60 must be 10.10.60.3, not .1 (TKT-2109, live capture showed .1).

Verification

Command	Where	Expected result
show vlan brief	All switches	VLANs 10–100 + 99 with the names in section 2; each access range in the VLAN from section 3; no user port left in VLAN 1 or 99.
show interfaces trunk	ASWs, L3SW1/2	Gi0/1–2 (ASW) and Gi1/0/3–8 (Dist) trunking, native 99, allowed list identical on both ends.
show interfaces status	ASWs	Used Fa0/x ports connected in the correct VLAN; Gi0/1–2 show 'trunk'. No err-disabled ports.
show spanning-tree summary / interface fa0/1 detail	ASWs	PortFast + BPDU Guard on Fa0/1–20 only; Gi0/1–2 are normal (non-edge) STP ports.
show etherchannel summary	L3SW1/2	Po1 flagged routed + in use (RU), LACP; Gi1/0/23 and Gi1/0/24 bundled (P).
show ip interface brief	L3SW1/2	Port-channel1 10.255.255.1 / .2 up/up; Vlan10–100 up/up with .2 (L3SW1) / .3 (L3SW2); no SVI owns .1.
ping 10.255.255.2	L3SW1	Replies across Po1.
show cdp neighbors	All switches	Each ASW sees L3SW1 on Gi0/1 and L3SW2 on Gi0/2; each Dist sees ASW1 on Gi1/0/3 ... ASW6 on Gi1/0/8.